

Lecture Notes · Geometry · Session 1 — The affine space: points, vectors, lines and planes

All the session material with the full formalization: the axioms of the affine space and their consequences, the three forms of the equation with their equivalence proved, and the theorem identifying affine subspaces with solution sets. Exercises and a Going further section at the end.

1.1 The affine space

Vector spaces have a privileged point: the 0. The plane of classical geometry does not — no point is special. What there *is* between two points is the **displacement** connecting them. The structure that captures this:

Definition 1.1 (affine space). An **affine space** over a vector space V (over K) is a nonempty set $\mathbb{A}$ of **points** together with an operation

$$\mathbb{A} \times V \rightarrow \mathbb{A}, \quad (P, v) \mapsto P + v,$$

satisfying: 1. $P + 0 = P$ for every point P ; 2. $(P + v) + w = P + (v + w)$ (translating twice is translating by the sum); 3. for each pair of points (P, Q) there exists a **unique** vector, denoted $\vec{PQ}$, with $P + \vec{PQ} = Q$.

The **dimension** of $\mathbb{A}$ is that of V . **Model:** $\mathbb{A}^n$ = the points of K^n , with $V = K^n$ acting by addition of tuples; the axioms are immediate.

Proposition 1.2 (consequences). In every affine space: 1. $\vec{PP} = 0$, and $P + v = P \Rightarrow v = 0$. 2. (**Chasles**) $\vec{PQ} + \vec{QR} = \vec{PR}$. 3. $\vec{QP} = -\vec{PQ}$. 4. (**parallelogram rule**) $\vec{PQ} = \vec{P'Q'} \iff \vec{PP'} = \vec{QQ'}$.

Proof. (1) $P + 0 = P$ by axiom 1, and the vector taking P to P is unique (axiom 3): $\vec{PP} = 0$; the second claim is the same uniqueness. (2) $P + (\vec{PQ} + \vec{QR}) = (P + \vec{PQ}) + \vec{QR} = Q + \vec{QR} = R$ (axiom 2), and by uniqueness that vector is $\vec{PR}$. (3) Chasles with $R = P$: $\vec{PQ} + \vec{QP} = \vec{PP} = 0$. (4) By Chasles twice: $\vec{PQ'} = \vec{PP'} + \vec{P'Q'} = \vec{PQ} + \vec{QQ'}$; equating $\vec{P'Q'} = \vec{PQ}$ amounts to equating $\vec{PP'} = \vec{QQ'}$ (cancelling in V). ■

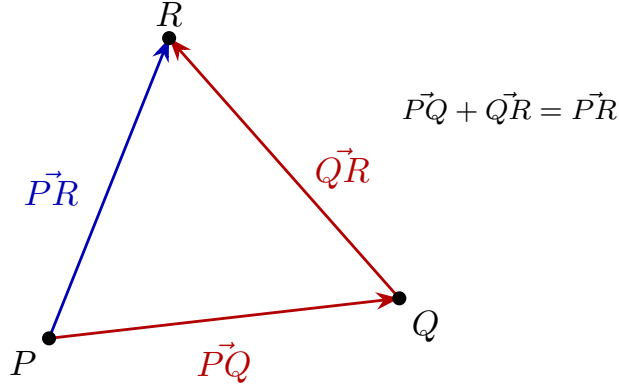


Figure 1: Chasles' relation: $\vec{PQ} + \vec{QR} = \vec{PR}$ (chained vectors), and $\vec{QP} = -\vec{PQ}$.

Affine = linear + reference point

Proposition 1.3. Once a point $O \in \mathbb{A}$ (an **origin**) is fixed, the assignment $P \mapsto \vec{OP}$ is a **bijection** between the points of $\mathbb{A}$ and the vectors of V .

Proof. Injective: if $\vec{OP} = \vec{OQ}$, then $P = O + \vec{OP} = O + \vec{OQ} = Q$. Surjective: $v = O(\vec{O} + v)$ by axiom 3. ■

The identification **depends on the choice of O** : with another origin O' , the vector associated with P changes ($O'P = O'O + \vec{OP}$, Chasles). That is why *point* and *vector* are different concepts: identifying them requires a choice. This is the precise content of the slogan “**affine = linear + reference point**”.

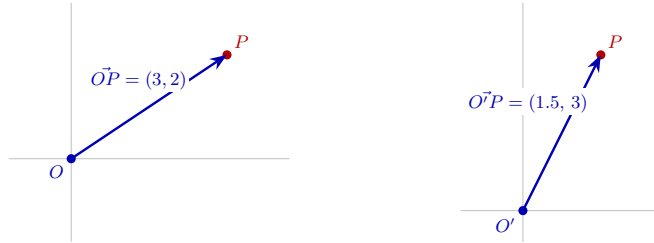


Figure 2: Drag the origin O (in the interactive version): the point P does not move, but its vector $\vec{OP}$ — and hence its coordinates — changes with the choice of O .

[see interactive version]

Definition 1.4 (frame of reference and coordinates). A **frame of reference** of $\mathbb{A}$ (dimension n) is a pair (origin O , basis $\{b_1, \dots, b_n\}$ of V). The **coordinates** of the point P are the coordinates of the vector $\vec{OP}$ in that basis — unique, by Proposition 3.18 of Linear Algebra.

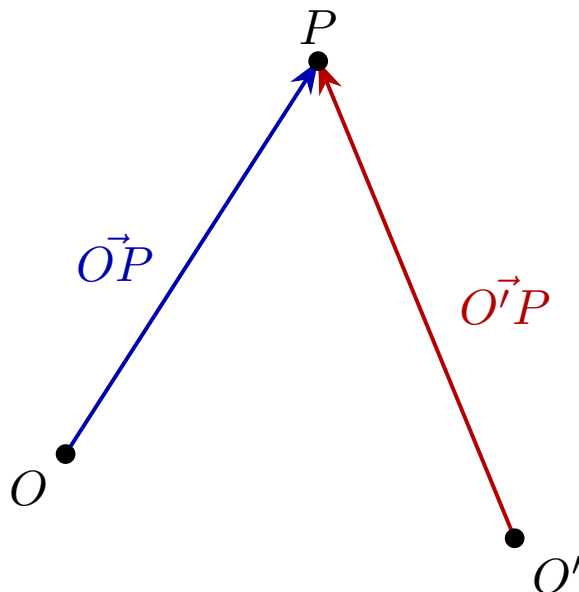


Figure 3: The base point is not intrinsic: the same P has different vectors $\vec{OP}$ and $\vec{O'P}$ depending on the origin, and hence different coordinates.

1.2 Affine subspaces; lines and planes

Definition 1.5. An **affine subspace** of $\mathbb{A}$ is a set of the form

$$P + W = \{ P + w : w \in W \},$$

with P a point and $W \subseteq V$ a vector subspace, called the **direction space**. Its **dimension** is $\dim W$: **points** ($\dim 0$), **lines** ($\dim 1$), **planes** ($\dim 2$), **hyperplanes** ($\dim n - 1$).

Proposition 1.6 (the direction space is intrinsic; the base point is not). If $P + W = P' + W'$, then $W = W'$ and $P' \in P + W$. Conversely, $P + W = P' + W$ for **any** $P' \in P + W$.

Proof. If $P' \in P + W$, then $\vec{PP'} \in W$ and $P' + W = P + (\vec{PP'} + W) = P + W$ (because $\vec{PP'} + W = W$: translating a subspace by one of its own vectors leaves

it unchanged — closure). For the first part: $W = \{\vec{XY} : X, Y \in P + W\}$ (differences of points of the set, via Chasles), which depends only on the set. ■

Definition 1.7 (line and plane, vector form). The **line** through P with **direction vector** $v \neq 0$: $r = P + \text{span}(v) = \{P + tv : t \in K\}$. The **plane** through P with **independent** direction vectors u, v : $\pi = P + \text{span}(u, v) = \{P + su + tv : s, t \in K\}$.

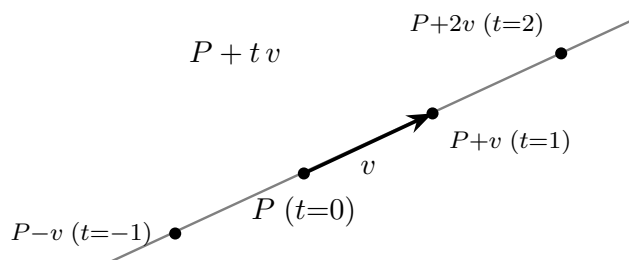


Figure 4: The line $P + tv$: as t varies, the whole line is traced out in the direction of the direction vector v (here $t = -1, 0, 1, 2$).

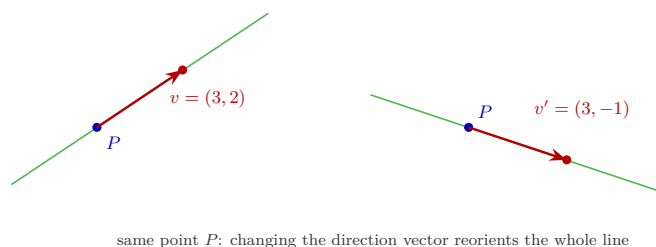


Figure 5: Interactive version: drag the tip of the direction vector to see how, as v changes, the whole line reorients.

[see interactive version]

The three forms and their equivalence

In coordinates (once a frame of reference is fixed in $\mathbb{A}^n$):

1. **Vector form:** $X = P + t_1 v_1 + \dots + t_k v_k$.
2. **Parametric:** the above, coordinate by coordinate (n equations with k parameters).
3. **Implicit:** the set as the solutions of a linear system $Ax = b$.

Theorem 1.8 (affine subspaces = solution sets). In $\mathbb{A}^n$: 1. The solution set of **any** consistent system $Ax = b$ is an affine subspace, with direction space

the kernel of A (and dimension $n - \text{rank} A$). 2. Conversely, **every** affine subspace $P + W$ is the solution set of some consistent linear system.

Proof. (1) It is the Theorem “general = particular + kernel” (Linear Algebra, Theorem 4.7) reread with Definition 1.5: solutions = $x_p + \ker A$; the dimension is Theorem 4.6 of Linear Algebra.

(2) It suffices to find a **homogeneous** system whose solutions are exactly W (of dimension k); afterwards the right-hand side is adjusted with $b := A \cdot P$ to translate to $P + W$ (the solutions of $Ax = AP$ are $P + \ker A$ by part 1). Let $\{w_1, \dots, w_k\}$ be a basis of W and M the $k \times n$ matrix with rows w_i . The kernel of M has dimension $n - k$ (the rank of M is k : independent rows); let $\{z_1, \dots, z_{n-k}\}$ be a basis of it and A the matrix with rows z_j . Then:

- each w_i is a solution of $Ax = 0$: the condition $Aw_i = 0$ says $\sum_l (z_j)_l (w_i)_l = 0$ for each j , which is exactly $Mz_j = 0$ read backwards (the expression is **symmetric** in w_i and z_j). Hence $W \subseteq \ker A$.
- $\dim(\ker A) = n - \text{rank} A = n - (n - k) = k = \dim W$. And a subspace contained in another of the same finite dimension coincides with it. *Indeed:* a basis of W is an independent family of k elements inside $\ker A$; by Lemma 3.18b(1) of Linear Algebra it extends to a basis of $\ker A$ — but that basis also has k elements ($\dim \ker A = k$), so the extension adds nothing: the basis of W was already a basis of $\ker A$, and $W = \text{span} = \ker A$. ■

Method 1.9 (converting between forms).

- *Parametric* $\rightarrow$ *implicit*: **eliminate the parameters** — treat the parametric equations as a system in the parameters and express the consistency condition in x (in practice: solve and substitute, or reduce to echelon form).
- *Implicit* $\rightarrow$ *parametric*: **solve the system** and parametrize — literally Method 4.4 of Linear Algebra.

Example 1.10. The line of $\mathbb{R}^3$ through $P = (1, 0, 2)$ with direction vector $v = (1, 1, -1)$:

- *Vector form*: $(x, y, z) = (1, 0, 2) + t(1, 1, -1)$.
- *Parametric*: $x = 1 + t$, $y = t$, $z = 2 - t$.
- *Implicit*: solving for $t = y$ and substituting: $x - y = 1$, $y + z = 2$ (system of rank 2; dimension of solutions $3 - 2 = 1$ ✓).

Hyperplanes. A single nondegenerate linear equation $a_1x_1 + \dots + a_nx_n = b$ defines an affine subspace of dimension $n - 1$ (rank 1): a **hyperplane**. Lines in the plane and planes in space are the cases $n = 2, 3$; the reading “each equation of a system carves out a hyperplane, and solving is intersecting them” (Session 1 of Linear Algebra) is now fully formalized.

Exercises

1. ★ For the line of $\mathbb{R}^3$ through $P = (0, 1, 1)$ and $Q = (2, 1, 3)$: find a direction vector, write the three forms and check that Q satisfies them.
2. ★ Convert to parametric form (by solving) the plane $x - 2y + z = 4$ of $\mathbb{R}^3$, and to implicit form (by eliminating parameters) the plane $(x, y, z) = (1, 0, 0) + s(1, 1, 0) + t(0, 1, 1)$.
3. ★ The system $\{x + y - z = 1; 2x - y = 0\}$ is consistent. Describe its solution set as an affine subspace: base point, direction space (a basis) and dimension.
4. ★ **(Point vs. vector / affine vs. linear.)** The line $y = x + 1$ of $\mathbb{R}^2$: is it a vector subspace? is it an affine subspace? Give its direction space and exhibit two different choices of base point (Proposition 1.6).
5. Prove with Chasles: the midpoint M of P and Q (defined by $\vec{PM} = \frac{1}{2}\vec{PQ}$, with $K = \mathbb{R}$) satisfies $\vec{MP} + \vec{MQ} = 0$.
6. Check the three axioms of Definition 1.1 in the model $\mathbb{A}^n$.
7. Let r be the line through P with direction vector v . Prove that $Q \in r \iff \vec{PQ}$ is proportional to v , and deduce a criterion for **three points to be collinear**.
8. In Theorem 1.8.2, follow the construction with $W = \text{span}((1, 1, 0), (0, 1, 1)) \subseteq \mathbb{R}^3$ and obtain implicit equations for the plane $(1, 2, 3) + W$.

Going further and references

Going further (affine combinations and barycenters). Although “adding points” makes no sense, the combinations $\sum \lambda_i P_i$ with $\sum \lambda_i = 1$ do (the result does not depend on the origin used to compute them): these are the **affine combinations**, and with them come barycenters, segments and convexity. See de Burgos, *Álgebra lineal y geometría cartesiana*, or Audin, *Geometry*, ch. I.

References. Axiomatic affine space: de Burgos; Audin, ch. I. Lines, planes and the forms of the equation: de Burgos; Hernández, *Álgebra y geometría*. Affine subspaces as solutions: Hefferon, One.I (implicitly), and the Linear Algebra notes, Session 4.

Lecture Notes · Geometry · Session 2 — Relative positions

The general method (intersecting = solving, classified by Rouché–Capelli), parallelism via directions, the tables for the plane and for space **deduced** case by case, the determinant criterion for skew lines and a complete example with a parameter. Exercises and a Going further section at the end.

2.1 The general method

Principle 2.1 (intersecting = solving). If two affine subspaces of $\mathbb{A}^n$ are given in implicit form, $\mathcal{F} = \{x : Ax = b\}$ and $\mathcal{G} = \{x : Cx = d\}$, their **intersection** is the solution set of the **joint system**

$$\begin{bmatrix} A \\ C \end{bmatrix} x = \begin{bmatrix} b \\ d \end{bmatrix},$$

which is classified with the **Rouché–Capelli theorem** (known in Spain as Rouché–Frobenius; Linear Algebra, Theorem 4.3): the intersection is empty if the ranks differ; if they coincide ($= r$), it is an affine subspace of dimension $n - r$ (Theorem 1.8). *If an object is given in parametric form, substitute it into the implicit equations of the other (usually the shortest route), or convert everything to implicit form (Method 1.9).*

Definition 2.2 (parallelism). Two affine subspaces are **parallel** if the direction space of one contains that of the other ($W \subseteq W'$ or $W' \subseteq W$). *(For two lines: proportional direction vectors; line and plane: the line’s direction vector belongs to the plane’s direction space.)*

Proposition 2.3. Two parallel affine subspaces are either disjoint, or one contains the other.

Proof. Suppose $W \subseteq W'$ and that $\mathcal{F} = P + W$ and $\mathcal{G} = Q + W'$ share a point R . By Proposition 1.6 we may take R as the base point of both: $\mathcal{F} = R + W \subseteq R + W' = \mathcal{G}$. ■

Parallelism is decided **without solving anything**: it is a **rank condition on the direction vectors** (is $\text{rank}(\text{all the direction vectors together})$ equal to the maximum of the separate ranks?).

2.2 The tables, deduced

Each case **is deduced** in three steps: (i) set up the joint system; (ii) bound the possible ranks; (iii) read off each combination with Rouché–Capelli and the dimension count $n - r$. The tables below are the record of that deduction.

Two lines in the plane ($n = 2$)

System of 2 equations (one per line), 2 unknowns; $1 \leq \text{rank} A \leq 2$.

$\text{rank} A$	$\text{rank}[A b]$	Reading	Position
2	2	consistent, $\dim = 0$	intersecting (one point)
1	1	consistent, $\dim = 1$	coincident
1	2	inconsistent	distinct parallel lines

(There are no more cases: $\text{rank}[A|b] \leq 2$ and it is never smaller than $\text{rank} A$.)

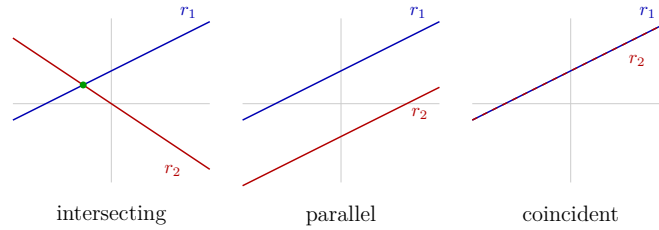


Figure 6: Relative positions of two lines in the plane (interactive version: drag the points of r_2): they meet at one point (intersecting), they do not meet (parallel) or they coincide.

[see interactive version]

In space ($n = 3$)

Two planes (2 equations): $\text{rank}A \leq 2 < 3$, so if there is an intersection its dimension is ≥ 1 :

Proposition 2.4. Two planes of $\mathbb{R}^3$ **never** meet in exactly one point.

Proof. The joint system has 2 equations and 3 unknowns: $\text{rank}A \leq 2$, and if it is consistent the dimension of the intersection is $3 - \text{rank}A \geq 1$. ■

$\text{rank}A$	$\text{rank}[A b]$	Position
2	2	they meet in a line
1	1	coincident
1	2	distinct parallel planes

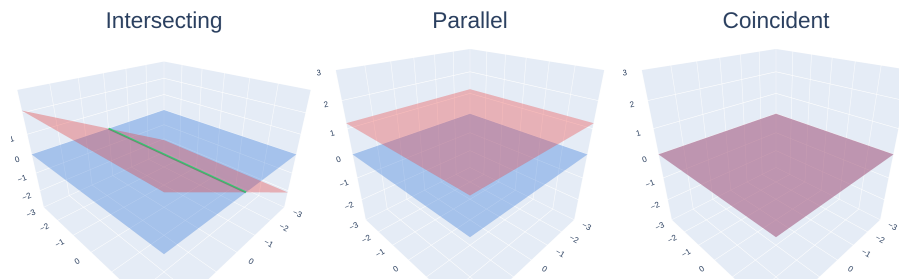


Figure 7: Relative positions of two planes in space: they meet in a line (intersecting), they are parallel or they coincide — with a slider running through the cases.

[see interactive version]

Line and plane ($2+1 = 3$ equations): cases (3,3) one **point**; (2,2) the line **contained** in the plane; (2,3) **parallel** (no contact). (*The case $\text{rank} A = 3$ inconsistent is impossible: with full rank the system is always consistent — $\text{rank}[A|b] \leq 3$.*)

Two lines ($2+2 = 4$ equations, 3 unknowns): (3,3) **intersecting** at a point; (2,2) **coincident**; (2,3) **distinct parallel lines**; and the novelty,

(3,4) : **skew** — they do not meet, *yet* they are not parallel.

Proposition 2.5 (determinant criterion for skew lines). Let $r = P + \text{span}(v)$ and $s = Q + \text{span}(w)$ be lines in $\mathbb{R}^3$. Then

$$r \text{ and } s \text{ are skew} \iff \det(\vec{PQ}, v, w) \neq 0,$$

(the three columns: the vector connecting the lines and the two direction vectors).

Proof. ($\Leftarrow$) If the determinant is nonzero, $\{\vec{PQ}, v, w\}$ is independent (Linear Algebra, Theorem 6.11). In particular v, w are not proportional (not parallel) and the lines do not meet: if they shared a point R , then $\vec{PQ} = \vec{PR} + \vec{RQ}$ would be a combination of v and w (since $\vec{PR} \parallel v$, $\vec{RQ} \parallel w$), against the independence. ($\Rightarrow$) Contrapositive: if the determinant is 0, the family is dependent. If v, w are proportional, the lines are parallel (not skew). If not, $\{v, w\}$ is independent and then $\vec{PQ} \in \text{span}(v, w)$ (in a nontrivial dependence, the coefficient of $\vec{PQ}$ cannot be 0, and one solves for it — field). Writing $\vec{PQ} = \alpha v + \beta w$, the point $P + \alpha v = Q + \beta w$ lies on both lines: they meet (not skew). ■

Why there are no skew lines in the plane: the criterion would require three independent vectors in K^2 , impossible (Linear Algebra, exercise 8 of Session 3: the dimension forbids it). The existence of skew lines is a **dimensional** phenomenon: room is needed.

Example 2.6 (a skew pair). $r : (0, 0, 0) + t(1, 0, 0)$ (the x -axis) and $s : (0, 0, 1) + t(0, 1, 0)$. With $\vec{PQ} = (0, 0, 1)$: $\det \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix} = 1 \neq 0$ — skew. (*Intuition: two corridors on different floors with perpendicular directions.*)

2.3 Example with a parameter (complete)

Example 2.7 (the three regimes). Position of $r : (x, y, z) = (0, 0, 1) + t(1, a^2 - 2, 0)$ and $\pi : x + y = a - 1$, according to $a \in \mathbb{R}$.

Substituting the points of the line, $(t, (a^2 - 2)t, 1)$, into the implicit equation:

$$t + (a^2 - 2)t = a - 1 \iff (a^2 - 1)t = a - 1.$$

A single equation in t , and its analysis is the trichotomy of Session 1 of Linear Algebra in geometric guise:

- $a \neq \pm 1$: nonzero coefficient $\rightarrow$ unique solution $t = \frac{a-1}{a^2-1} = \frac{1}{a+1}$: the line **meets** the plane at one point.
- $a = 1$: it becomes $0 \cdot t = 0$, which **always** holds: the line is **contained** in the plane. (Check: direction $(1, -1, 0)$ and plane $x + y = 0$; the points $(t, -t, 1)$ all satisfy it.)
- $a = -1$: it becomes $0 \cdot t = -2$, which **never** holds: line and plane **parallel** (no contact).

Note the mechanics: the regime is decided by **two different things** — whether the coefficient vanishes (parallelism of direction) and, in that case, whether or not the constant term vanishes (contained vs. strictly parallel). That is why the parameter had to appear on both sides, with zeros at different values of a .

Exercises

1. ★ Analyze the relative position, setting up the system and comparing ranks:
 - (a) the lines of $\mathbb{R}^2$: $x + y = 1$ and $2x + 2y = 3$;
 - (b) the planes $x - y + z = 0$ and $2x - 2y + 2z = 5$;
 - (c) the lines of $\mathbb{R}^3$: $r : (1, 0, 0) + t(1, 1, 0)$ and $s : (0, 1, 0) + t(0, 1, 1)$ — also apply the determinant criterion (Proposition 2.5).
2. ★ Find the intersection of $\pi_1 : x + y + z = 3$ and $\pi_2 : x - y = 1$, parametrized, and check that its dimension agrees with $3 - \text{rank}$.
3. ★ Analyze according to a : the line $r : (x, y, z) = (0, 0, 1) + t(1, a, 0)$ and the plane $\pi : x + y = 2$. **Only two** of the three possible regimes appear — identify them, give the critical value of a , and **explain why the case “line contained in the plane” is impossible for every a** (hint: what two things would have to vanish at once, and can they?).
4. ★ Prove that the lines $r : (0, 0, 0) + t(1, 1, 0)$ and $s : (1, 0, 1) + t(1, 1, 1)$ are skew, and give implicit equations for each.
5. Deduce the line–plane table in $\mathbb{R}^3$: list the possible rank pairs of the joint system and justify why $(3, \text{inconsistent})$ does not appear.
6. Prove: if two lines of $\mathbb{R}^3$ meet at **two** distinct points, they are coincident. (Hint: two points determine the direction.)
7. (Conceptual.) Explain why Definition 2.2 of parallelism includes the case “one contained in the other”, and why that is consistent with Proposition 2.3.

Going further and references

Going further (pencils of planes). The planes containing a given line $r = \pi_1 \cap \pi_2$ are exactly those with equation $\lambda(\text{eq. of } \pi_1) + \mu(\text{eq. of } \pi_2) = 0$ with $(\lambda, \mu) \neq (0, 0)$: the **pencil of planes** with

axis r . Useful for imposing “a plane containing r and satisfying X ” without parametrizing. See de Burgos.

References. Relative positions via ranks: de Burgos, *Álgebra lineal y geometría cartesiana*; Hernández, *Álgebra y geometría*. Skew lines and the determinant criterion: de Burgos. The underlying analysis of systems: the Linear Algebra notes, Session 4.

Lecture Notes · Geometry · Session 3 — Euclidean space: dot product, distances and angles

The full metric structure: the dot product and its properties proved, Cauchy–Schwarz with proof (and from it the well-defined angle and the triangle inequality), Pythagoras, the orthogonal projection derived and the distance formulas worked out. The counterfactual of why this structure demands $\mathbb{R}$, resolved in detail. Exercises and a Going further section at the end.

3.1 The dot product

Until now the geometry was **affine**: incidence, parallelism, dimensions. To speak of lengths and angles an additional structure is needed, and a single operation brings it.

Definition 3.1. The (standard) **dot product** of $u, v \in \mathbb{R}^n$:

$$u \cdot v = u_1v_1 + u_2v_2 + \cdots + u_nv_n.$$

Proposition 3.2 (properties). For $u, v, w \in \mathbb{R}^n$ and $a, b \in \mathbb{R}$: 1. **Symmetry:** $u \cdot v = v \cdot u$. 2. **Bilinearity:** $(au + bw) \cdot v = a(u \cdot v) + b(w \cdot v)$ (and, by symmetry, likewise in the second argument). 3. **Positivity:** $u \cdot u \geq 0$, with equality **if and only if** $u = 0$.

Proof. (1) and (2): straight from the definition (commutativity and distributivity of $\mathbb{R}$, term by term). (3) $u \cdot u = u_1^2 + \cdots + u_n^2$ is a sum of real squares, ≥ 0 ; and a sum of real squares is 0 only if **each** summand is 0, that is, $u = 0$. ■

Remark 3.3 (where $\mathbb{R}$ is used — the thread of fields, final twist). Property (3) is the one that does **not survive** in an arbitrary field:

- In $K = \mathbb{Z}/5\mathbb{Z}$, with $u = (1, 2)$: $u \cdot u = 1 + 4 = 5 = 0$, with $u \neq 0$. “Positivity” cannot even be stated ($\mathbb{Z}/5\mathbb{Z}$ is not ordered) and its consequence fails outright.
- In $\mathbb{C}$ with the same formula: $u = (1, i)$ gives $u \cdot u = 1 + i^2 = 0$. (That is why in $\mathbb{C}$ a different operation is used, with conjugates — see Going further.)

- Moreover, the **norm** (below) needs square roots of nonnegative numbers, which $\mathbb{R}$ guarantees.

Conclusion: **affine** geometry works over any field K (Sessions 1–2); **Euclidean** geometry — this session — lives over $\mathbb{R}$.

3.2 Norm, distance and the key inequality

Definition 3.4. The **norm** of u : $\|u\| = \sqrt{u \cdot u}$ (well defined by 3.2.3). The **distance** between two points: $d(P, Q) = \|\vec{PQ}\|$. (In the plane, $\|(a, b)\| = \sqrt{a^2 + b^2}$: the Pythagorean theorem turned into a definition.)

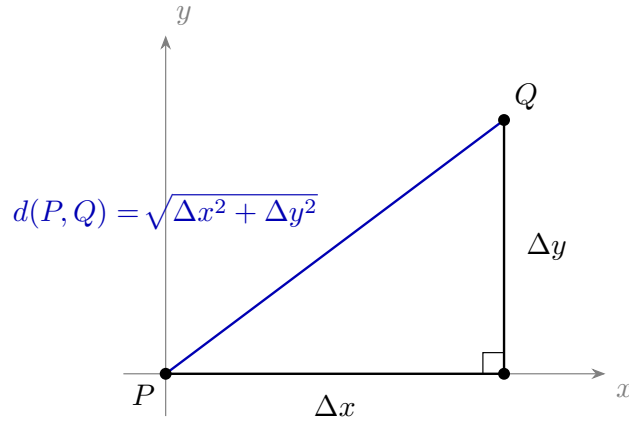


Figure 8: Distance in the plane via Pythagoras: $d(P, Q) = \sqrt{\Delta x^2 + \Delta y^2}$, the hypotenuse of the triangle with legs $\Delta x, \Delta y$.

Immediate properties: $\|u\| \geq 0$ with equality only at $u = 0$; $\|\lambda u\| = |\lambda| \|u\|$ (a λ^2 comes out of the bilinearity and the square root returns it as $|\lambda|$).

Theorem 3.5 (Cauchy–Schwarz inequality). For all $u, v \in \mathbb{R}^n$:

$$|u \cdot v| \leq \|u\| \|v\|,$$

with equality **if and only if** u and v are proportional.

Proof. If $v = 0$, both sides are 0 (and $0 = 0 \cdot u$: proportional). Let $v \neq 0$. For every $t \in \mathbb{R}$, positivity gives

$$0 \leq \|u - tv\|^2 = (u - tv) \cdot (u - tv) = \|u\|^2 - 2t(u \cdot v) + t^2\|v\|^2,$$

(using bilinearity and symmetry). The right-hand side is a quadratic polynomial in t with leading coefficient $\|v\|^2 > 0$ that is never negative: its **discriminant** is ≤ 0 :

$$4(u \cdot v)^2 - 4\|v\|^2\|u\|^2 \leq 0 \iff |u \cdot v| \leq \|u\|\|v\|.$$

Equality: the discriminant is 0 exactly when the polynomial has a root t_0 , that is, $\|u - t_0 v\| = 0$, that is (3.2.3), $u = t_0 v$: proportional. Conversely, if $u = t_0 v$ the equality is checked directly. ■

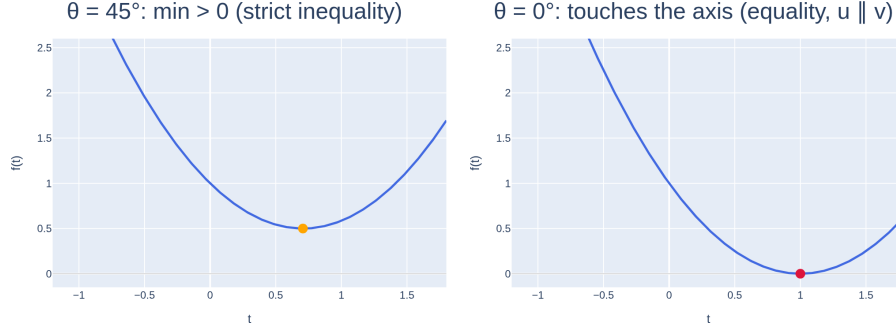


Figure 9: Cauchy–Schwarz, made visible: $f(t) = \|u - tv\|^2$ is a parabola that never dips below the axis; its discriminant being ≤ 0 is the inequality, and it touches the axis (equality) only if u, v are proportional.

[see interactive version]

Corollary 3.6 (triangle inequality). $\|u + v\| \leq \|u\| + \|v\|$.

Proof. Expanding and bounding the cross term,

$$\|u + v\|^2 = \|u\|^2 + 2(u \cdot v) + \|v\|^2 \leq \|u\|^2 + 2\|u\|\|v\| + \|v\|^2 = (\|u\| + \|v\|)^2,$$

using $u \cdot v \leq |u \cdot v| \leq \|u\|\|v\|$ (Cauchy–Schwarz); conclude by taking square roots (both sides are ≥ 0). ■

3.3 Angle and orthogonality

The goal is to **measure the angle** between two vectors. In the plane one has $u \cdot v = \|u\|\|v\| \cos \theta$, so the cosine is the **route** to recovering the angle; we take it as the definition in $\mathbb{R}^n$, and Cauchy–Schwarz guarantees that it makes sense.

Definition 3.7. For $u, v \neq 0$, the **angle** $\theta \in [0, \pi]$ between them is the unique one with

$$\cos \theta = \frac{u \cdot v}{\|u\| \|v\|}.$$

Why it is well defined (the point usually omitted): by Cauchy–Schwarz, the quotient lies in $[-1, 1]$, which is exactly the range of the cosine on $[0, \pi]$, where the cosine is injective. Without Theorem 3.5, the definition would not even make sense. Rewritten: $u \cdot v = \|u\|\|v\| \cos \theta$ — the dot product measures “how much u points in the direction of v ” (positive: acute; zero: right; negative: obtuse).

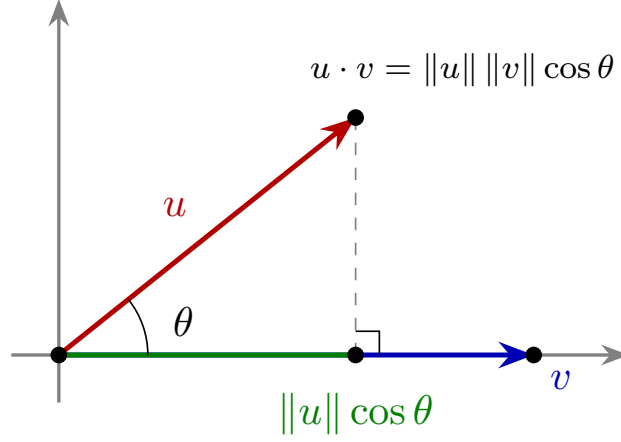


Figure 10: The dot product as a projection: $u \cdot v = \|u\| \|v\| \cos \theta$, with $\|u\| \cos \theta$ the shadow of u on v .

Definition 3.8. $u \perp v$ (**orthogonal**) if $u \cdot v = 0$.

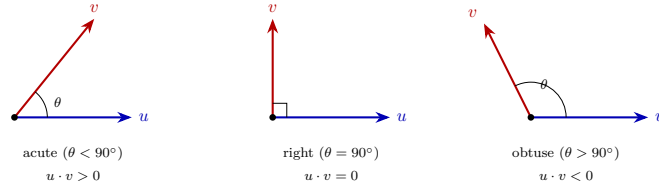


Figure 11: The sign of $u \cdot v$ according to the angle: acute (> 0), right ($= 0$, orthogonal), obtuse (< 0).

Proposition 3.9. 1. (**Pythagoras**) If $u \perp v$, then $\|u + v\|^2 = \|u\|^2 + \|v\|^2$. 2. For fixed $v \neq 0$, the set $\{v\}^\perp = \{u : u \cdot v = 0\}$ is a **subspace** of $\mathbb{R}^n$, of dimension $n - 1$ (a vector hyperplane).

Proof. (1) Expanding with bilinearity and symmetry, $\|u + v\|^2 = (u + v) \cdot (u + v) = \|u\|^2 + 2(u \cdot v) + \|v\|^2$; if $u \perp v$ the cross term $2(u \cdot v)$ is 0. (2) It is the kernel of the row matrix $v^\top$ (the condition $u \cdot v = 0$ is one homogeneous linear equation): a subspace of dimension $n - \text{rank}(v^\top) = n - 1$ (Linear Algebra, Theorem 4.6). ■

The orthogonal complement of v is a plane through the origin

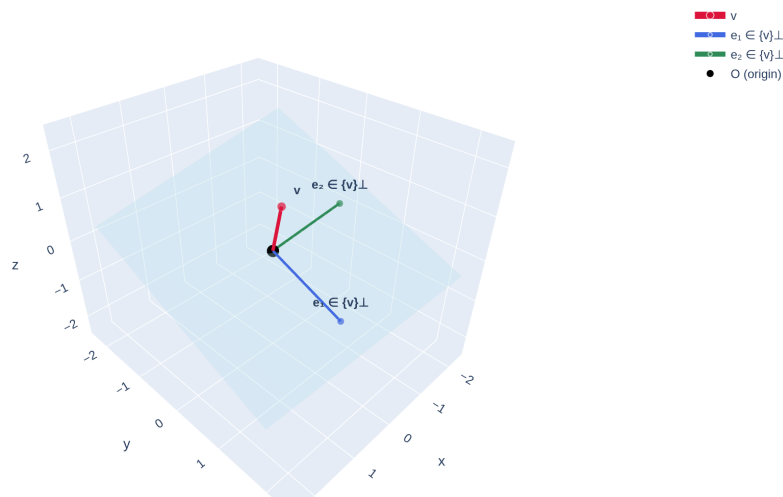


Figure 12: The orthogonal complement of a nonzero vector v is a plane through the origin: $\{v\}^\perp = \{u : u \cdot v = 0\}$. It is the basis of the idea of the “normal” of a plane.

[see interactive version]

3.4 Orthogonal projection and distances

Proposition 3.10 (orthogonal projection, derived). Given u and $v \neq 0$, there exists a unique multiple of v , called the **orthogonal projection of u onto v** , such that the remainder is perpendicular to v ; and it equals

$$\text{proj}_v(u) = \frac{u \cdot v}{\|v\|^2} v.$$

Proof. We look for t with $(u - tv) \perp v$:

$$(u - tv) \cdot v = 0 \iff u \cdot v - t\|v\|^2 = 0 \iff t = \frac{u \cdot v}{\|v\|^2},$$

(solvable because $\|v\|^2 \neq 0$). The condition determines t uniquely, so the projection is unique. ■

The formula **falls out of the geometric condition in one line**.

The resulting decomposition $u = \text{proj}_v(u) + (u - \text{proj}_v(u))$ splits u into “part parallel to v ” + “part perpendicular”.

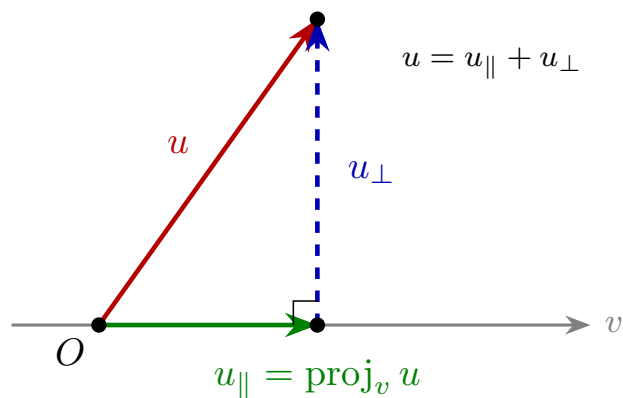


Figure 13: Orthogonal decomposition: $u = u_{\parallel} + u_{\perp}$, with $u_{\parallel} = \text{proj}_v u$ the component along v and $u_{\perp}$ the perpendicular one.

Method 3.11 (distance from a point to a line). For Q and the line $r = P + \text{span}(v)$: decompose $\vec{PQ}$ with the projection onto v ; the distance is the norm of the **perpendicular part**:

$$d(Q, r) = \|\vec{PQ} - \text{proj}_v(\vec{PQ})\|.$$

(That this really is the minimum of $d(Q, X)$ over $X \in r$ follows from Pythagoras: for any other point of the line, the squared distance adds the perpendicular part — fixed — plus a nonzero parallel part. Details: exercise 6.)

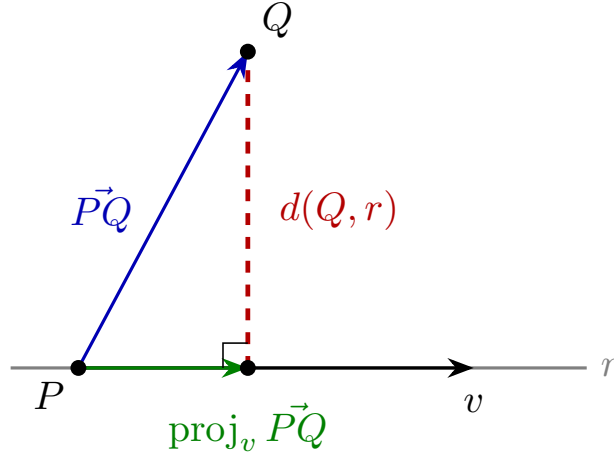


Figure 14: Distance from Q to the line r : the length of the component of $\vec{PQ}$ perpendicular to the direction v .

Proposition 3.12 (the normal of a plane, proved). In $\mathbb{R}^3$, the vector $n = (a, b, c)$ is **orthogonal to all the directions** of the plane $\pi : ax + by + cz = d$.

Proof. The equation says exactly $n \cdot X = d$ for the points X of the plane (dot product against the coordinate vector). If $X, Y \in \pi$, subtracting: $n \cdot \vec{XY} = n \cdot Y - n \cdot X = d - d = 0$. Every direction of the plane is of the form $\vec{XY}$: orthogonal to n . ■

Method 3.13 (distance from a point to a plane). For Q and $\pi : ax + by + cz = d$ with normal n : take any $P \in \pi$ and project $\vec{PQ}$ onto n — the part parallel to the normal is what separates Q from the plane:

$$d(Q, \pi) = \|\text{proj}_n(\vec{PQ})\| = \frac{|n \cdot \vec{PQ}|}{\|n\|} = \frac{|a q_1 + b q_2 + c q_3 - d|}{\sqrt{a^2 + b^2 + c^2}},$$

where the last equality uses $n \cdot \vec{PQ} = n \cdot Q - n \cdot P = (a q_1 + b q_2 + c q_3) - d$ (Proposition 3.12). The point–plane distance formula is thus **derived** in two lines: projection + the reading of the plane’s equation.



[see interactive version]

and use Definition 3.7.)

8. True or false? “If $u \cdot v = u \cdot w$ with $u \neq 0$, then $v = w$.” Justify or give a counterexample (in $\mathbb{R}^3$!).

Going further and references

Going further (general inner products). The three properties of Proposition 3.2 are taken as **axioms** and define the *inner product spaces*, where this whole section reproduces word for word (Cauchy–Schwarz included: the proof of 3.5 only used the axioms — check it). In $\mathbb{C}^n$ the right product is $\sum u_i \overline{v_i}$ (with conjugates, to rescue positivity). See Axler, *Linear Algebra Done Right*, ch. 6.

Going further (least squares). The orthogonal projection onto a subspace (not just a line) solves the problem of “the best approximate solution” of an inconsistent system — the **least squares** regression of data science. Strang, §4.2–4.3.

Optional going further (distances in general — metric spaces). The Euclidean distance is one case of a broader concept. A **distance** (or *metric*) on a set X is a function $d : X \times X \rightarrow \mathbb{R}_{\geq 0}$ with: (i) $d(x, y) = 0 \iff x = y$; (ii) **symmetry**, $d(x, y) = d(y, x)$; (iii) the **triangle inequality**, $d(x, z) \leq d(x, y) + d(y, z)$. The Euclidean one satisfies them (the triangle inequality is Corollary 3.6). There are other useful ones: the taxicab distance $\sum_i |x_i - y_i|$ or the maximum distance $\max_i |x_i - y_i|$. Further reading: **W. A. Sutherland, *Introduction to Metric and Topological Spaces* (Oxford University Press)**, ch. 2; it will reappear in later Analysis/Topology courses. Not assessed.

References. Dot product and Euclidean geometry: Hernández, *Álgebra y geometría*; de Burgos. Cauchy–Schwarz and inner products: Axler, ch. 6.A. Projections: Strang, §4.2.

Lecture Notes · Geometry · Session 4 — Cross product, areas and volumes

The closing session of the module, with everything proved: the properties of the cross product (including Lagrange’s identity and from it the interpretation as an area), the scalar triple product identified with the determinant, the coplanarity tests, and the distance between skew lines derived. The “symbolic determinant” mnemonic justified. Exercises and a Going further section at the end.

4.1 The cross product in $\mathbb{R}^3$

The geometry of space constantly calls for a vector **perpendicular to two others** (normals of planes, above all). The cross product constructs it explicitly.

Definition 4.1. For $u = (u_1, u_2, u_3)$, $v = (v_1, v_2, v_3) \in \mathbb{R}^3$:

$$u \times v = (u_2v_3 - u_3v_2, \quad u_3v_1 - u_1v_3, \quad u_1v_2 - u_2v_1) \in \mathbb{R}^3.$$

Remark 4.2 (the mnemonic, justified). The “symbolic determinant” rule

$$u \times v = \text{”det”} \begin{pmatrix} e_1 & e_2 & e_3 \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{pmatrix}$$

is not a definition (it mixes vectors and scalars in one matrix), but it works as a computation rule: expanding formally along the first row (cofactors, Sarrus’ rule by blocks), the three “coefficients” of e_1, e_2, e_3 that come out are exactly the three components of Definition 4.1 — check it (exercise 7). Its real merit appears in Proposition 4.6.

Proposition 4.3 (algebraic properties). 1. **Antisymmetry:** $v \times u = -(u \times v)$; in particular $u \times u = 0$. 2. **Bilinearity:** linear in each argument. 3. $u \times v = 0 \iff u, v$ linearly dependent (proportional).

Proof. (1) and (2): direct componentwise check (each component is a bilinear antisymmetric expression in the pairs of coordinates). (3) ($\Leftarrow$) If $v = \lambda u$, each component is of the type $u_i(\lambda u_j) - u_j(\lambda u_i) = 0$ (or $u = 0$, trivial). ($\Rightarrow$) Let M be the 2×3 matrix with rows u, v . If they are independent, $\text{rank} M = 2$: reducing to echelon form leaves **two pivots**, in certain columns $j < k$. Consider the 2×2 submatrix of columns j, k — a **minor**. Row operations on M act **on entire rows**, so they induce exactly row operations on that submatrix; in the echelon form, the submatrix of the pivot columns is triangular with nonzero diagonal, that is, invertible — and since row operations preserve invertibility (they are multiplication by elementary matrices, Linear Algebra 2.15.2 and 3.24), the **original** submatrix is invertible: its determinant, nonzero (Linear Algebra, Theorem 6.8). But the three 2×2 minors of M are, up to sign, **the three components of $u \times v$** (compare with Definition 4.1): some component is nonzero and $u \times v \neq 0$. ■

Proposition 4.4 (perpendicularity). $(u \times v) \perp u$ and $(u \times v) \perp v$.

Proof. Direct:

$$(u \times v) \cdot u = (u_2v_3 - u_3v_2)u_1 + (u_3v_1 - u_1v_3)u_2 + (u_1v_2 - u_2v_1)u_3 = 0,$$

because the six terms cancel in pairs ($u_1u_2v_3$ with $-u_2u_1v_3$, etc.). Likewise with v . ■

Theorem 4.5 (Lagrange’s identity and the area).

$$\|u \times v\|^2 = \|u\|^2\|v\|^2 - (u \cdot v)^2, \quad \text{and therefore} \quad \|u \times v\| = \|u\|\|v\|\sin\theta,$$

which is the **area of the parallelogram** with sides u and v .

Proof. The identity is a computation: expanding $\|u \times v\|^2$ (the sum of the squares of the three components) and $\|u\|^2\|v\|^2 - (u \cdot v)^2$ (products of sums), both sides turn out to be the same sum of terms $u_i^2 v_j^2$ ($i \neq j$) minus cross terms $2u_i u_j v_i v_j$ ($i < j$) — the term-by-term verification is mechanical and is left indicated (exercise 9, with guidance). From there, using $u \cdot v = \|u\|\|v\| \cos \theta$ (Session 3):

$$\|u \times v\|^2 = \|u\|^2\|v\|^2(1 - \cos^2 \theta) = \|u\|^2\|v\|^2 \sin^2 \theta,$$

and since $\theta \in [0, \pi]$, $\sin \theta \geq 0$ and square roots can be taken with no loss. The interpretation: base $\|u\|$ times height $\|v\| \sin \theta$. ■

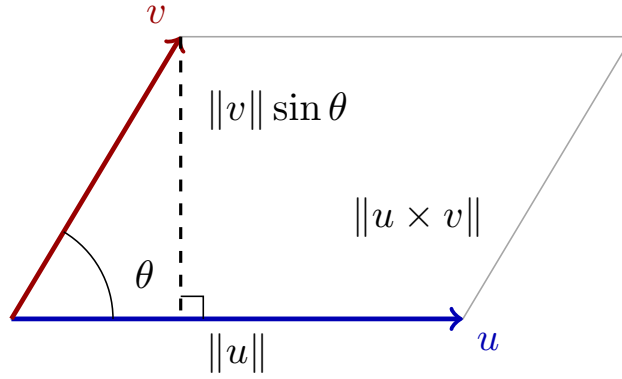


Figure 16: Parallelogram with sides u, v : base $\|u\|$ and height $\|v\| \sin \theta$ (the component of v perpendicular to u); its area is $\|u \times v\|$.

The “ $u \times v = 0$ ” case is a limiting case, not a separate property. Proposition 4.3.3 is exactly Theorem 4.5 in the limit: $\|u \times v\| = \|u\|\|v\| \sin \theta = 0 \iff \sin \theta = 0 \iff u \parallel v$ — the parallelogram flattens and the area drops to zero ($u \times u = 0$ is the extreme case $\theta = 0$). The algebraic proof of 4.3.3 with minors remains useful (it uses no trigonometry), but conceptually it *is* the area hitting zero.

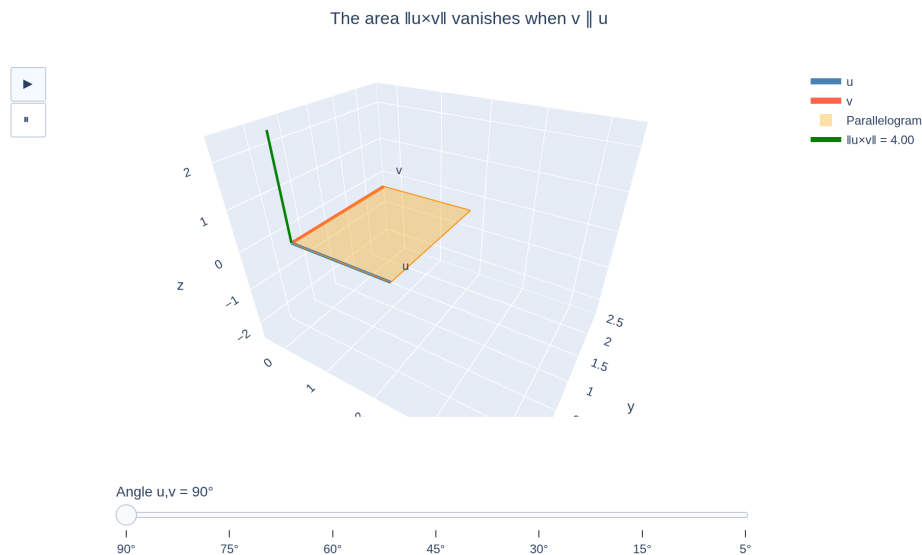


Figure 17: The area $\|u \times v\|$ vanishes as v becomes parallel to u : the parallelogram flattens and $u \times v \rightarrow 0$.

[see interactive version]

Orientation: of the two possible perpendicular directions, $u \times v$ points according to the **right-hand rule** — the cross product *sees* the orientation, in line with the sign of the determinant (Linear Algebra, Session 6). Antisymmetry ($v \times u = -u \times v$) is that sensitivity made algebra.

4.2 Scalar triple product and volumes

Definition and Proposition 4.6 (scalar triple product = determinant). The **scalar triple product** of $u, v, w \in \mathbb{R}^3$ is $[u, v, w] = u \cdot (v \times w)$, and it coincides with the determinant:

$$[u, v, w] = \det \begin{pmatrix} u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \\ w_1 & w_2 & w_3 \end{pmatrix}.$$

Proof. Expanding the determinant **along the first row** (cofactors):

$$\det = u_1(v_2w_3 - v_3w_2) - u_2(v_1w_3 - v_3w_1) + u_3(v_1w_2 - v_2w_1),$$

which is exactly $u_1(v \times w)_1 + u_2(v \times w)_2 + u_3(v \times w)_3 = u \cdot (v \times w)$ (Definition 4.1). (Here the mnemonic of 4.2 shows its truth: the “symbolic determinant” is this expansion with e_i in place of u_i .) ■

Theorem 4.7 (volumes). $|[u, v, w]|$ is the **volume of the parallelepiped** with edges u, v, w . Consequently (Linear Algebra, Theorem 6.11):

$$u, v, w \text{ coplanar (dependent)} \iff [u, v, w] = 0.$$

Proof (volume). **Degenerate case first:** if v, w are dependent, then $v \times w = 0$ (Proposition 4.3.3), the “base” has area 0, the parallelepiped is flattened (volume 0) and the scalar triple product equals $u \cdot 0 = 0$: the equality holds trivially. **Case v, w independent:** then $n = v \times w \neq 0$ and one can proceed. Base: the parallelogram of v, w , with area $\|n\|$ (Theorem 4.5). Height: the component of u perpendicular to the base, which is the projection of u onto the normal n (Proposition 4.4): height $= \|\text{proj}_n u\| = \frac{|u \cdot n|}{\|n\|}$ (Session 3 — the division is legitimate: $\|n\| \neq 0$). Volume = base $\times$ height $= \|n\| \cdot \frac{|u \cdot n|}{\|n\|} = |u \cdot (v \times w)|$. (Fully consistent with “the determinant measures volumes” from Linear Algebra S6 — here we have rebuilt it from metric geometry.) ■

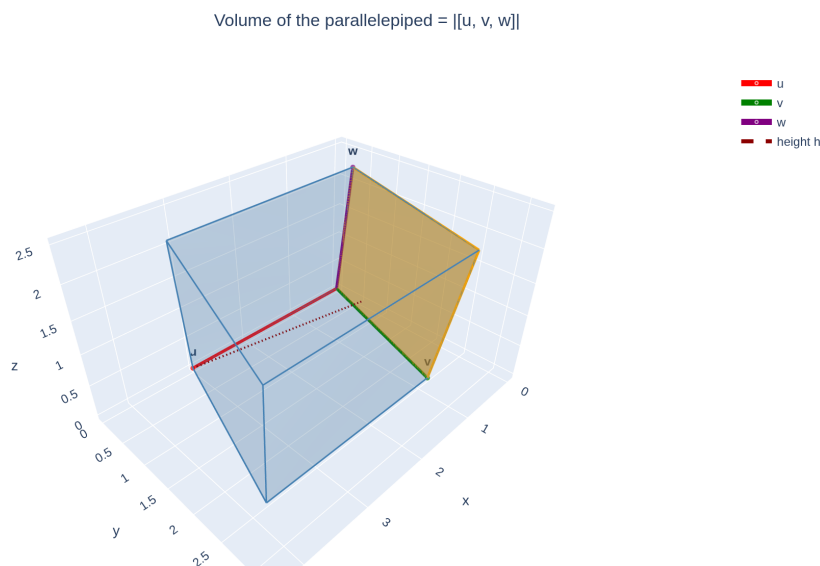


Figure 18: Parallelepiped with edges u, v, w from a common vertex: base the parallelogram of v, w (area $\|v \times w\|$) and height $\|\text{proj}_n u\|$ with $n = v \times w$; if coplanar, it flattens onto a plane (volume 0).

[see interactive version]

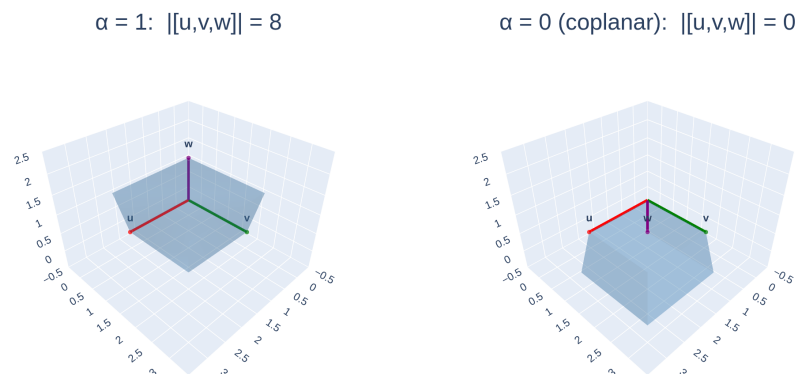


Figure 19: Coplanarity $\iff$ volume 0: as the third vector approaches the plane of the other two, the parallelepiped flattens and $[u, v, w] \rightarrow 0$.

[see interactive version]

Immediate computational applications: area of a triangle ($\frac{1}{2}\|u \times v\|$), volume of a tetrahedron ($\frac{1}{6}[u, v, w]$ — for reference), collinearity and coplanarity tests for points (form the difference vectors and look at the rank/determinant).

4.3 Applications

Method 4.8 (the normal and the implicit equation of the plane, in one line). Plane through P with direction vectors u, v : its **normal** is $n = u \times v$ (perpendicular to both, 4.4, and nonzero by 4.3.3 since they are independent). The implicit equation is then

$$n \cdot \vec{PX} = 0 \quad \text{that is} \quad n_1x + n_2y + n_3z = n \cdot P,$$

(a point X lies on the plane exactly when $\vec{PX}$ is a direction, that is, perpendicular to the normal — Proposition 3.12 in reverse).

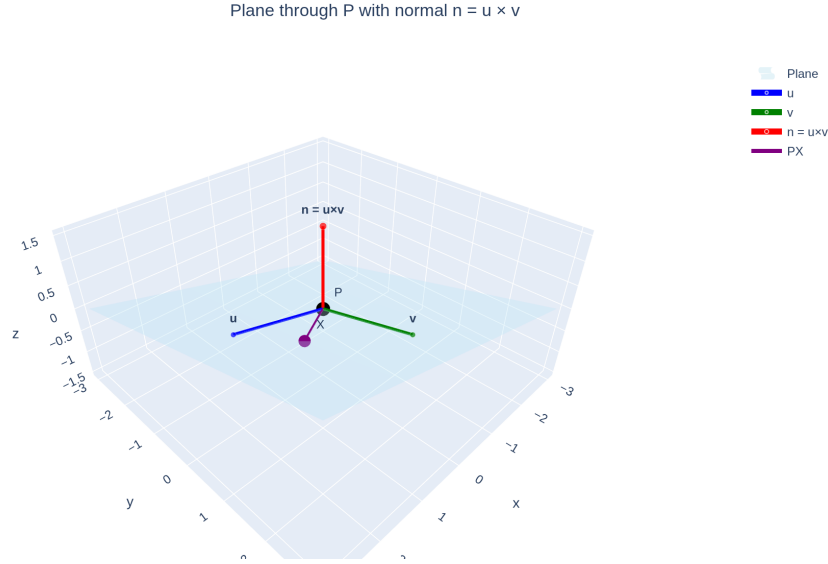


Figure 20: Plane through P with direction vectors u, v and the normal $n = u \times v$ sticking out of the plane; a generic point X with $\vec{PX}$ contained in the plane (perpendicular to n).

[see interactive version]

Theorem 4.9 (distance between skew lines). Let $r = P + \text{span}(v)$ and $s = Q + \text{span}(w)$ be skew. Then

$$d(r, s) = \frac{|[\vec{PQ}, v, w]|}{\|v \times w\|}.$$

Proof. Since r, s are skew, v, w are independent and $n = v \times w \neq 0$ is **perpendicular to both directions**. For any points $X = P + \alpha v \in r$ and $Y = Q + \beta w \in s$:

$$\vec{XY} = \vec{PQ} - \alpha v + \beta w \implies \vec{XY} \cdot n = \vec{PQ} \cdot n$$

(the terms in v and w die against n). That is: **the component along the common normal is the same for every pair of points**, and equals $\frac{|\vec{PQ} \cdot n|}{\|n\|}$ upon normalizing. No distance $\|\vec{XY}\|$ can be smaller than its component along n (projection, Session 3), so $d(r, s) \geq \frac{|\vec{PQ} \cdot n|}{\|n\|}$; and that bound **is attained** by choosing α, β that kill the components of $\vec{XY}$ perpendicular to n (possible: $\{v, w, n\}$ is a basis of $\mathbb{R}^3$, and the coordinates of $\vec{PQ}$ along v, w are compensated with α, β). Finally $\vec{PQ} \cdot n = \vec{PQ} \cdot (v \times w) = [\vec{PQ}, v, w]$ (Definition 4.6). ■

Note the piece-by-piece assembly: skew lines (S2, determinant criterion), normal (cross product), projection (S3), scalar triple product (4.6). **The formula is the whole module condensed.**

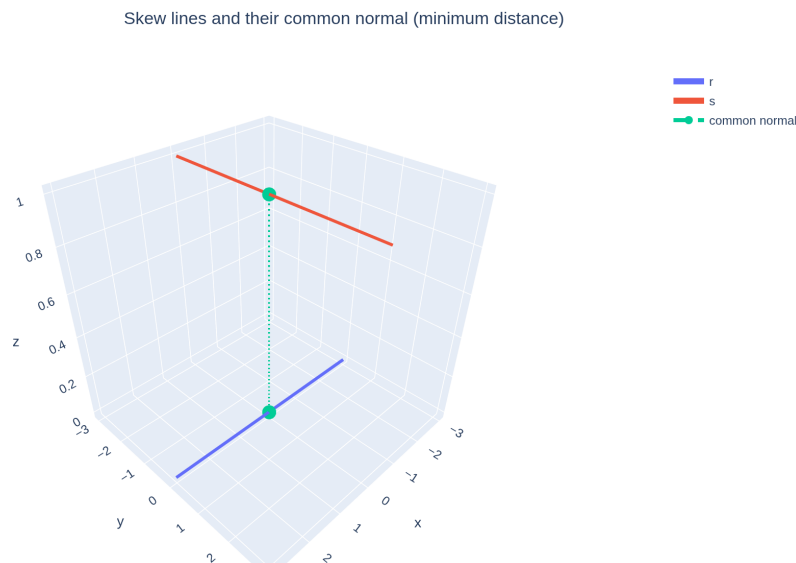


Figure 21: Two skew lines r, s and the segment of the common normal realizing the minimum distance; rotating the view shows that they do not meet.

[see interactive version]

Exercises

1. ★ With $u = (1, 0, 1)$ and $v = (0, 1, 1)$: compute $u \times v$, check the perpendicularity with the dot product, and find the **area** of the parallelogram and of the triangle they determine.
2. ★ Find the implicit equation of the plane through $P = (1, 1, 0)$ with direction vectors $u = (1, 0, 1)$, $v = (0, 1, 1)$, via the normal (Method 4.8). Check with a second point of the plane.
3. ★ Compute the volume of the parallelepiped with edges $(1, 0, 0)$, $(1, 1, 0)$, $(1, 1, 1)$. Then decide: (a) whether the **vectors** $(1, 2, 3)$, $(2, 3, 4)$, $(3, 4, 5)$ — with a common origin — are coplanar (scalar triple product test); (b) whether the **four points** $(0, 0, 0)$, $(1, 2, 3)$, $(2, 3, 4)$, $(3, 4, 5)$ are coplanar (*form the difference vectors from the first one — and note that three points are always coplanar: that is why the test for points needs four*).
4. ★ Compute the **distance between the skew lines** of Example 2.6 (r : the x -axis; $s : (0, 0, 1) + t(0, 1, 0)$) with Theorem 4.9, and check that the result matches the intuition of the picture (“corridors on different floors”).
5. Verify the antisymmetry $v \times u = -(u \times v)$ with Definition 4.1 and connect it with the determinant property “swapping rows changes the sign” (via 4.6).
6. Show that $u \times v$ is **not associative**: find u, v, w with $(u \times v) \times w \neq u \times (v \times w)$.

(Hint: try vectors from the standard basis.)

7. Check that the formal expansion of the “symbolic determinant” (Remark 4.2) along the first row reproduces Definition 4.1.
8. Follow the minors argument of the proof of 4.3.3 with $u = (1, 2, 4)$ and $v = (2, 4, 9)$: check that they are independent, compute the **three** 2×2 minors and observe that the first one (columns 1,2) **vanishes** and yet $u \times v \neq 0$ — the argument guarantees *some* nonzero minor, not the first one.
9. **(Lagrange, guided.)** Expand $\|u \times v\|^2$ and $\|u\|^2\|v\|^2 - (u \cdot v)^2$ and check that both equal $\sum_{i < j} (u_i v_j - u_j v_i)^2$.
10. **(Module synthesis.)** For $r : (1, 0, 0) + t(1, 1, 0)$ and $s : (0, 1, 1) + t(1, -1, 1)$: decide their position (S2), and if they are skew compute their distance (4.9); if not, whichever distance applies (S3).

Going further and references

Going further (why only in $\mathbb{R}^3$?). A bilinear antisymmetric product that returns a *vector in the same space*, perpendicular to the factors and with the norm of the area, exists only in dimension 3 (and, surprisingly, in dimension 7, via the octonions). In general dimension, the natural object is not a vector but the determinant/alternating forms — the cross product is a happy accident of $\mathbb{R}^3$. Introductory reference: Stillwell, *Naïve Lie Theory*, ch. 1, or Massey’s classic survey, *Cross products of vectors in higher dimensional Euclidean spaces*.

Going further (isometries — the module’s additional material). The **isometries** are the transformations that preserve distances: translations, rotations, reflections and their compositions. The linear ones have matrices with orthonormal columns, and their determinant is ± 1 : $+1$ preserves orientation (rotations), -1 reverses it (reflections) — the closing clasp with Session 6 of Linear Algebra. See Audin, *Geometry*, ch. II–III, or de Burgos.

References. Cross product and scalar triple product: Hernández, *Álgebra y geometría*; de Burgos. Volumes and determinants: Strang, §5.3; the Linear Algebra notes, Session 6. Distance between skew lines: de Burgos.